



BACHARELADO EM
CIÊNCIA DA COMPUTAÇÃO

**LEVERAGING LARGE LANGUAGE MODELS AND
SYNTHETIC SOCIETIES FOR SCALABLE SIMULATION
OF SOCIAL BEHAVIOR AND ASSESSMENT OF
FIDELITY IN COMPLEX SOCIAL SYSTEMS**

LUCAS TOURINHO MAMEDE

Brasília - DF, 2025

LUCAS TOURINHO MAMEDE

**LEVERAGING LARGE LANGUAGE MODELS AND
SYNTHETIC SOCIETIES FOR SCALABLE SIMULATION
OF SOCIAL BEHAVIOR AND ASSESSMENT OF
FIDELITY IN COMPLEX SOCIAL SYSTEMS**

Trabalho de Conclusão de Curso apresentado como requisito parcial para a obtenção de grau de Bacharel em Ciência da Computação, pelo Instituto Brasileiro de Ensino, Desenvolvimento e Pesquisa (IDP).

Orientador

Me. Klayton Rodrigues de Castro

Brasília - DF, 2025

LUCAS TOURINHO MAMEDE

**LEVERAGING LARGE LANGUAGE MODELS AND
SYNTHETIC SOCIETIES FOR SCALABLE SIMULATION
OF SOCIAL BEHAVIOR AND ASSESSMENT OF
FIDELITY IN COMPLEX SOCIAL SYSTEMS**

Trabalho de Conclusão de Curso apresentado como requisito parcial para a obtenção de grau de Bacharel em Ciência da Computação, pelo Instituto Brasileiro de Ensino, Desenvolvimento e Pesquisa (IDP).

Aprovado em 05/12/2025

Banca Examinadora

Me. Klayton Rodrigues de Castro- Orientador

Me. Patrícia da Silva de Oliveira- Examinadora interna

Me. Lucas Maurício Castro e Martins- Examinador interno

ABSTRACT

The rapid expansion of Large Language Models (LLMs) has advanced research in reasoning, decision-making, and computational social simulation. However, the field still lacks a rigorous, scientifically grounded framework for constructing synthetic societies whose emergent behaviors align with empirical socioeconomic patterns. While LLM agents can emulate individual cognition, there is no principled methodology for grounding these agents in real-world information, constraining their collective behavior, or assessing the empirical fidelity of the resulting synthetic populations. This gap highlights a critical methodological limitation at the intersection of artificial intelligence, computational social science, and multi-agent systems. This research addresses this limitation by proposing a **Behavioral Grounding Framework** that formalizes the foundations for building, structuring, and evaluating LLM-based synthetic populations. The framework supports rigorous integration of socioeconomic data into agent profiles, enforces structured multi-agent interactions with clear behavioral assumptions, and establishes quantitative criteria for validating simulation outcomes. By providing a transparent, modular, and reproducible foundation, this work aims to advance the scientific use of LLM-based synthetic societies in economic modeling, policy analysis, and the study of collective behavior.

Keywords: synthetic societies, LLM Agents, behavioral grounding, empirical fidelity, socioeconomic modeling.

RESUMO

A rápida expansão dos Modelos de Linguagem Grande (LLMs) tem impulsionado a pesquisa em raciocínio, tomada de decisão e simulação social computacional. Contudo, a área ainda carece de um arcabouço rigoroso e cientificamente fundamentado para a construção de sociedades sintéticas cujos comportamentos emergentes se alinhem com padrões socioeconômicos empíricos. Embora os agentes baseados em LLM possam emular a cognição individual, não existe uma metodologia baseada em princípios para fundamentar esses agentes em informações do mundo real, restringir seu comportamento coletivo ou avaliar a fidelidade empírica das populações sintéticas resultantes. Essa lacuna evidencia uma limitação metodológica crítica na interseção entre inteligência artificial, ciência social computacional e sistemas multiagentes. Esta pesquisa aborda tal limitação propondo um **Arcabouço de Fundamentação Comportamental** que formaliza os fundamentos para a construção, estruturação e avaliação de populações sintéticas baseadas em LLMs. O arcabouço suporta a integração rigorosa de dados socioeconômicos nos perfis dos agentes, impõe interações multiagentes estruturadas com premissas comportamentais claras e estabelece critérios quantitativos para a validação dos resultados da simulação. Ao fornecer uma base transparente, modular e reprodutível, este trabalho visa avançar o uso científico de sociedades sintéticas baseadas em LLMs na modelagem econômica, na análise de políticas e no estudo do comportamento coletivo.

Palavras-chave: sociedades sintéticas, agentes LLM, fundamentação comportamental, fidelidade empírica, modelagem socioeconômica.

LIST OF FIGURES

1	Methodological structure adopted in this research.....	5
2	The Transformer model architecture.....	9
3	Comparison between Reason-Only, Act-Only, and ReAct (Reason + Act) agent paradigms.....	10
4	Structural transition from regular to random networks. The Small-World model (center) exhibits high clustering like regular lattices but short characteristic path lengths like random graphs, facilitating realistic social diffusion.....	11

LIST OF TABLES

1	Comparison of Related Works and the Proposed Approach.....	13
2	Strategic Schedule for Research Development	16

CONTENTS

1	Introduction	2
	1.1 Contextualization	2
	1.2 Problem Delimitation	3
	1.3 Justification	4
	1.4 Objectives	4
	1.5 Methodological Approach	5
	1.6 Structure of the work	5
2	Literature Review	8
	2.1 Theoretical Framework	8
	2.1.1 <i>The Era of Rules: Symbolic and Cognitive Architectures</i>	8
	2.1.2 <i>The Era of Optimization: Multi-Agent Reinforcement Learning</i>	9
	2.1.3 <i>The Generative Turn: From Foundation to Action</i>	9
	2.1.4 <i>Population Synthesis and Network Topology</i>	10
	2.2 Related Works	11
	2.2.1 <i>Capabilities</i>	11
	2.2.2 <i>The Reliability Crisis</i>	11
	2.3 Validation Framework and Metrics	12
	2.3.1 <i>Distributional Metrics</i>	12
	2.3.2 <i>Economic Inequality Metrics</i>	12
	2.4 Gap Analysis	13
3	Conclusion	15
	3.1 Final Considerations and Planning	15
	3.2 Future Work and Schedule	15
	References	17



1

1

INTRODUCTION

1.1 CONTEXTUALIZATION

Social science has long suffered from a singular, fatal constraint: the inability to run controlled experiments on history itself. We cannot collapse an economy just to test a hypothesis on inflation, nor can we deploy a radical social policy merely to observe the fallout. Synthetic Societies dismantle this barrier. By orchestrating thousands of autonomous, cognitively complex agents into a living *in silico* mirror of our world, this technology grants us the unprecedented power to simulate the future before it happens. It transforms the speculative art of forecasting into an experimental science, providing a “sandboxed reality” where stakeholders—from central banks to marketing strategists—can stress-test their most volatile interventions against a population that negotiates, adapts, and evolves, all without risking a single real-world consequence.

Within this scenario, Large Language Models (LLMs) transformed the field by overcoming the historical limitation of rigid agent thinking [1]. LLMs opened the possibility for agents to exhibit more flexible and human-aligned reasoning patterns, directly enabling synthetic societies capable of complex emergent behavior. This shift is central to the current research agenda, as it explains why LLM-based agents can serve as cognitively rich components in large-scale social simulations.

It is important to mention that even before the emergence of LLM-based agents, attempts to simulate social dynamics relied primarily on traditional Agent-Based Modeling (ABM). Classical artificial societies captured important macropatterns—such as segregation, mobility, and economic exchange—through rule-based agents [2, 3]. However, these agents operated under rigid, predefined behavioral rules and simplified utility functions [4], which prevented them from representing the cognitive richness, heterogeneity, and adaptive reasoning observed in real populations. This historical limitation establishes the conceptual inflection point that motivates the transition toward cognitively enhanced, LLM-driven agents.

These advances allow the construction of synthetic societies populated by LLM agents displaying heterogeneous backgrounds, beliefs, and economic profiles [5]. Researchers can now build digital laboratories to test economic policies, model social conflicts, and study collective decision-making with greater realism, capturing deeper

heterogeneity in systemic outcomes.

However, a major scientific gap remains regarding calibration and structural validity. LLM agent simulations are often developed with a focus on narrative plausibility rather than statistical fidelity. As Horton [6] highlights, while LLMs can act as simulated economic agents, their utility for the social sciences depends entirely on their ability to replicate structural empirical regularities rather than linguistic coherence. Currently, the field relies on ad hoc prompt designs that lack standardized mechanisms for directly injecting real-world census data or socioeconomic distributions into the agent generation pipeline. Without a rigorous methodology to ground these agents in empirical data—transforming them from “stochastic parrots” into calibrated personas—simulation outcomes risk becoming hallucinatory artifacts rather than reproducible social insights. This research aims to bridge this gap.

Therefore, the core scientific challenge is not the construction of LLM-driven synthetic societies, but their methodological grounding. Without empirically calibrated personas and validated collective patterns, synthetic populations risk producing plausible narratives rather than scientifically reliable results. This study positions itself precisely at this inflection point by proposing a methodological framework to transform LLM-based synthetic societies from exploratory artifacts into empirically anchored scientific instruments. This grounding perspective defines the methodological frontier of Synthetic Society research and directly motivates the problem addressed in this work.

Such a methodological foundation is relevant to researchers in computational social science, public policy analysts, economists, and any domain where population-level simulations depend on empirically grounded synthetic agents rather than narrative consistency alone.

1.2 PROBLEM DELIMITATION

This research addresses the absence of a robust, empirically grounded framework capable of using demographic datasets, such as census or survey data, as well as a reproducible pipeline for building, calibrating, and testing LLM-based synthetic societies. Although the technology to construct LLM agents already exists, there is still no established method to ensure that their collective behavior aligns with real-world socioeconomic patterns, systematically translates demographic datasets into agent personas, or validates whether the aggregate behavior of these agents reflects observed empirical regularities.

Moreover, existing studies rarely distinguish narrative plausibility from structural fidelity, making it difficult to determine whether simulated outcomes emerge from meaningful social mechanisms or from uncontrolled linguistic artifacts inherent to LLMs. The lack of standardized grounding procedures also undermines reproducibility, as current approaches depend on ad hoc prompt engineering and non-transparent calibration

heuristics.

Within this context, this work delimits its scope to proposing a fundamental methodology: creating agent personas directly from real data, defining consistent interaction rules, and establishing metrics to measure behavioral accuracy. The goal is to articulate the scientific standards required for valid applications and to define the minimal conceptual structure that enables future empirical implementations.

1.3 JUSTIFICATION

The lack of a rigorous method threatens the scientific legitimacy of LLM-based computational social science. For instance, if policy recommendations derived from synthetic societies rely on uncalibrated agents that do not reflect actual income distributions, policymakers may implement interventions that are ineffective or even detrimental when applied to real populations. Without standardized grounding in real data—such as income distributions, educational levels, and regional demographics—findings from synthetic societies cannot be replicated and may lead to misleading interpretations, limiting their value for research and policy [7].

The relevance of this work is grounded in advancing synthetic societies from exploratory demonstrations to empirically calibrated simulations capable of supporting analytical studies. Treating agents not as fictional characters but as statistical representations of real demographic groups provides the necessary foundation for structured analyzes of collective behavior and socioeconomic dynamics.

Researchers increasingly require methods that move beyond ad hoc prompt engineering toward reproducible and transparent procedures. Likewise, public institutions and organizations that consider using synthetic societies for forecasting, evaluation of economic measures, or scenario testing need models supported by measurable fidelity rather than narrative coherence alone.

This research is timely because current approaches still lack methodological consistency and clear validation mechanisms. By focusing on how to ground agent construction and assess collective behavior against real-world baselines, this study establishes a conceptual foundation that enables future empirical implementations while maintaining rigor and reproducibility.

1.4 OBJECTIVES

Large Language Models to construct synthetic societies, enabling scalable simulation of social behavior and quantitative assessment of fidelity in complex social systems. The specific goals are:

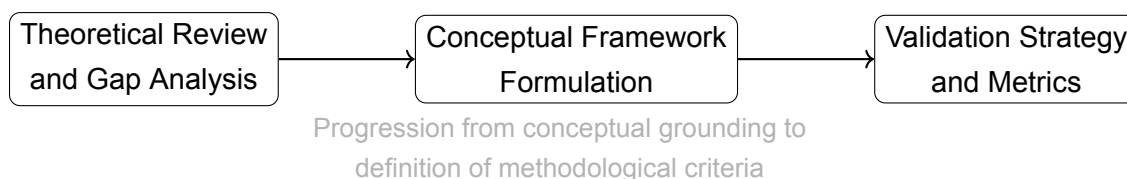
- To systematize the conceptual foundations required to transform demographic and socioeconomic datasets into semantically coherent LLM-based agent personas that compose synthetic societies;

- To define the architectural principles and behavioral constraints that structure the proposed Behavioral Grounding Framework, specifying how LLM-based agents can support scalable simulations of collective behavior;
- To discuss statistical distance measures and behavioral consistency criteria that could be employed to evaluate the fidelity of synthetic societies with respect to real social and economic patterns, framing acceptable fidelity thresholds as conceptual references rather than empirical goals for this stage;
- To synthesize a reproducible conceptual pipeline integrating data grounding, persona construction, interaction modeling, and fidelity assessment, forming the methodological basis for future empirical implementations.

1.5 METHODOLOGICAL APPROACH

To achieve these objectives, this work adopts a constructive and theoretically driven methodological approach, structured in three integrated movements, as summarized in Figure 1. First, a focused literature review is conducted on agent-based modeling, cognitive and generative architectures, synthetic populations, and validation metrics, in order to delineate the conceptual and methodological limitations of existing approaches and to extract the design requirements for the framework. Second, based on this theoretical consolidation, a Behavioral Grounding Framework is proposed, formally specifying the stages of data grounding, persona construction, interaction modeling, and definition of quantitative indicators of fidelity and inequality, always anchored in empirical socioeconomic distributions. Finally, a validation strategy is outlined in which synthetic societies generated under this framework are, in principle, assessed through distributional distance measures and inequality indices with respect to real-world baselines, establishing a coherent and reproducible route for subsequent empirical implementation without detailing operational choices that lie beyond the scope of this phase of the research.

Figure 1: Methodological structure adopted in this research.



Source: Own elaboration.

1.6 STRUCTURE OF THE WORK

This work is organized into the following chapters. After the Introduction, Chapter 2 reviews the literature, explains the theoretical framework, and analyzes related studies. Chapter 3 concludes with Final Considerations and planning for the development of the next steps.

2

2

LITERATURE REVIEW

This chapter establishes the theoretical and methodological foundations of the research. Following the principle of scientific ontogeny, it analyzes the evolution of agent-based simulation paradigms, tracing the path from symbolic architectures and cognitive models to modern generative approaches. It incorporates the fundamentals of computational population synthesis to justify the data grounding mechanisms. Finally, it critically reviews the state of the art in Large Language Model (LLM) agents, using recent cautionary studies to identify a methodological gap in empirical grounding and statistical validation.

2.1 THEORETICAL FRAMEWORK

The pursuit of autonomous agents capable of simulating human social behavior has evolved through distinct eras. Each era addressed the limitations of its predecessors while introducing new challenges.

2.1.1 The Era of Rules: Symbolic and Cognitive Architectures

Computational Social Science has historically relied on Agent-Based Modeling (ABM), a subset of Multi-Agent Systems (MAS), to study emergent social phenomena [2]. In MAS, multiple interacting intelligent agents coexist in a shared environment, creating complex dynamics that single-agent systems cannot replicate [8]. Early agents were modeled using the Belief-Desire-Intention (BDI) architecture. While BDI offered a logical framework for rational behavior, formalized through modal logics [9], it suffered from "cognitive brittleness": agents could not handle situations not explicitly foreseen by the designer. Research in this domain advanced from simple implementations to complex axiomatizations, such as BDICTL and Wooldridge's Logic Of Rational Agents (LORA) [10]. Early implementations, such as IRMA [11] and PRS [12], attempted to operationalize this logic but were computationally constrained by the need for manual rule specification.

Parallel efforts to build unified cognitive architectures, such as SOAR [13] and ACT-R [14], attempted to model general intelligence through symbolic production rules. However, these systems faced significant scalability constraints [4], as every social norm or concept had to be manually encoded. As detailed in Section 2.1.3, the emer-

gence of Large Language Models addresses precisely these limitations by providing agents with intrinsic, pre-trained semantic knowledge of the world.

2.1.2 The Era of Optimization: Multi-Agent Reinforcement Learning

To overcome the rigidity of manual rules, the field shifted towards Multi-Agent Reinforcement Learning (MARL). In this paradigm, agents learn optimal policies through trial-and-error, maximizing cumulative rewards [15].

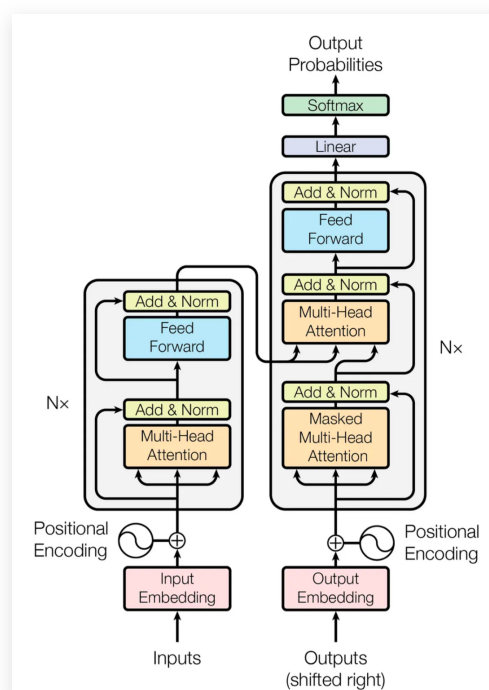
While MARL enabled breakthroughs in strategic coordination [16], Zhang et al. [17] highlights significant limitations for social simulation. These systems are often opaque “black boxes” and suffer from extreme sample inefficiency. Purely utility-driven agents also tend to exhibit behavior misaligned with human social norms, prioritizing reward maximization [18].

2.1.3 The Generative Turn: From Foundation to Action

The shift to generative agents resulted from technological advances in semantic processing and agency.

- Foundational Model (Transformers): The paradigm shift began with the Transformer architecture [19]. By utilizing self-attention mechanisms, Transformers enabled the training of models on vast corpora, providing agents with intrinsic “world knowledge” [20], as illustrated in Figure 2. This significantly mitigates reliance on manual rule specification, which had limited earlier architectures like SOAR.

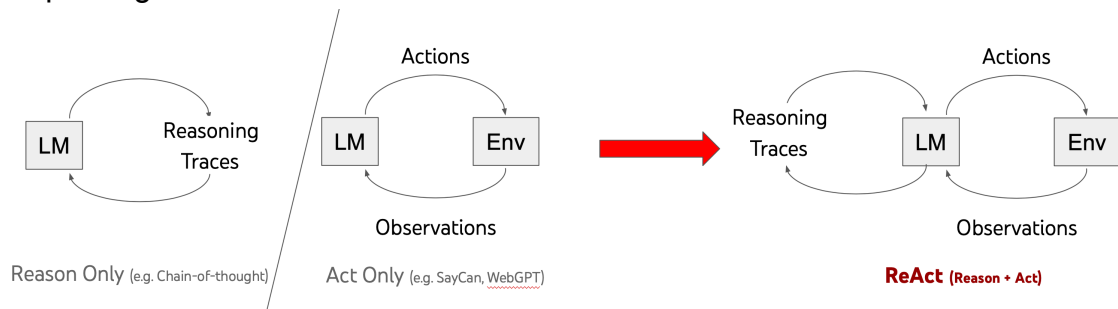
Figure 2: The Transformer model architecture.



Source: Adapted from Vaswani et al. [19].

- Reasoning (Chain-of-Thought): Wei et al. [21] discovered that prompting models to generate intermediate reasoning steps (Chain-of-Thought) activates emergent reasoning abilities. This allows agents to simulate cognitive processes rather than merely pattern-matching.
- Retrieval-Augmented Generation (RAG): To address the limitations of static training data and hallucinations, Lewis et al. [22] proposed RAG. This technique retrieves relevant external information (e.g., database records) and injects it into the model's context window during inference. In this thesis, RAG serves as the fundamental mechanism for the “Behavioral Grounding” of agents using Census microdata.
- Agency (ReAct): Reasoning alone is insufficient for situated simulation; agents must interact with their environment. Yao et al. [23] introduced the ReAct (Reason + Act) framework, creating a “perception-action loop” where the agent reasons, acts, observes the output, and updates its cognition. This mechanism is the theoretical basis for the interaction framework proposed in this thesis, as illustrated in Figure 3.

Figure 3: Comparison between Reason-Only, Act-Only, and ReAct (Reason + Act) agent paradigms.



Source: Adapted from ReAct Project Website [24].

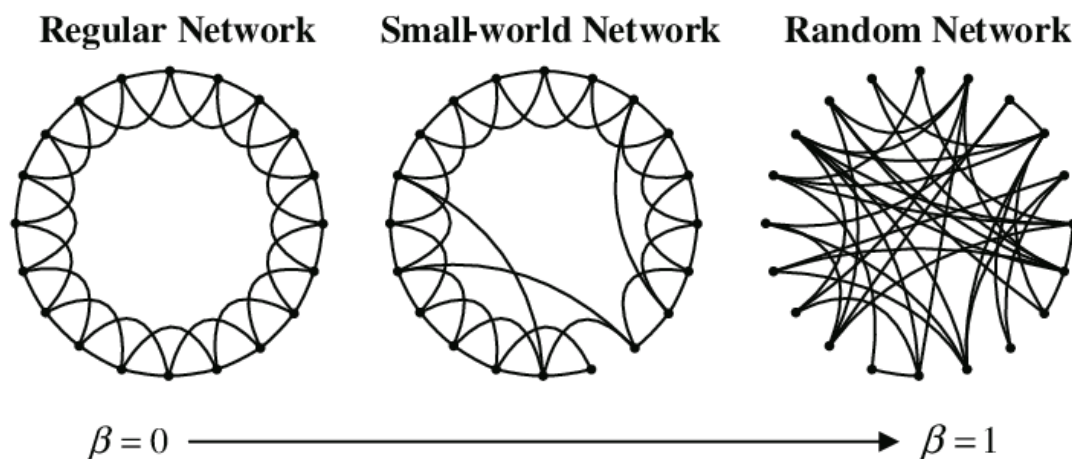
2.1.4 Population Synthesis and Network Topology

Simulating a realistic society requires more than intelligent agents; it requires a representative population and a plausible environment.

- Population Synthesis: Classic computational demography relies on techniques like Iterative Proportional Fitting (IPF) to generate synthetic populations that match aggregate census data [25]. However, traditional methods generate only static attributes. This thesis extends the concept by using LLMs to convert synthetic demographic data into semantic personas, a process called Behavioral Grounding.

- **Network Topology:** The structure of interaction dictates social diffusion. Theoretical grounding relies on the Small-World Network model [26], which shows that human social networks exhibit high clustering and short path lengths. Adhering to this topology prevents artifactual randomness in the economic exchanges modeled in this work, as illustrated in Figure 4.

Figure 4: Structural transition from regular to random networks. The Small-World model (center) exhibits high clustering like regular lattices but short characteristic path lengths like random graphs, facilitating realistic social diffusion.



Source:

Adapted from Watts and Strogatz [26].

2.2 RELATED WORKS

Current research explores the intersection of LLMs and social science, but recent studies highlight reliability issues with consistency and variance.

2.2.1 Capabilities

Generative Agents and Silicon Sampling Park et al. [1] validated the architecture for memory and reflection in a simulated town, "Smallville", demonstrating that agents could coordinate events. In a subsequent study scaling to 1,000 agents, Park et al. [27] further explored emergent community dynamics. However, these agents remained fictional and lacked empirical socioeconomic grounding. Concurrently, Argyle et al. [5] proposed "Silicon Sampling," demonstrating that LLMs trained on demographic profiles could reproduce voting patterns observed in the American National Election Studies (ANES). Aher et al. [28] expanded this to replicate classic psychological experiments. While promising, these approaches are static and lack the emergent properties of dynamic interaction.

2.2.2 The Reliability Crisis

Despite these successes, recent literature warns against uncritical reliance on LLMs such as Drift and Variance.

- **Persona Drift:** Li et al. [29] demonstrated that LLM personas exhibit systematic biases and behavioral drift that deviate from real-world behavioral patterns, failing to maintain consistency. This justifies the need for the proposed Memory Stream and Grounding mechanisms to enforce behavioral constraints.
- **Variance Underestimation:** Bisbee et al. [30] found that synthetic populations systematically underestimate the heterogeneity of real populations. LLMs tend to converge on "average" behaviors, failing to capture the long-tail diversity needed to study inequality.
- **Strategic Failure:** Gao et al. [31] showed that LLM agents fail to reproduce human equilibria in game-theoretic scenarios such as "Scylla Ex Machina", often deviating due to safety filters or prompt sensitivity.

2.3 VALIDATION FRAMEWORK AND METRICS

To address the reliability crisis, this research adopts a Validation and Verification approach as described by Collins et al. [32], focusing on empirical validation and distributional alignment.

2.3.1 Distributional Metrics

Comparing synthetic societies requires statistical distance measures.

- **KL Divergence:** Defined by MacKay [33], the Kullback-Leibler divergence is the canonical measure for comparing probability distributions. However, it fails when distributions have non-overlapping supports.
- **Wasserstein Distance:** As demonstrated by Arjovsky et al. [34] and Manzan [35], the Wasserstein Distance (Earth Mover's Distance) provides a meaningful geometric measure of the "work" required to transform the synthetic distribution into the empirical one. This geometric property makes it superior for validating continuous socioeconomic variables, which serve as the primary validation metric for this work.

2.3.2 Economic Inequality Metrics

To ground the simulation in socioeconomic reality, we use established inequality metrics defined by Cowell [36], specifically the Gini Coefficient and the Lorenz Curve. These serve as ground-truth patterns against which the emergent simulation outcomes are tested.

2.4 GAP ANALYSIS

Table 1 synthesizes the architectural evolution and identifies the multidimensional gap this thesis addresses, comparing the proposed framework with key related works.

Table 1: Comparison of Related Works and the Proposed Approach

Work	Architecture	Data Source	Critical Limitation	Validation
Rao and Georgeff [9]	Symbolic BDI	Manual Rules	No learning or multi-agent adaptation	Logic proof
Sutton and Barto [15]	RL	Reward signal	Limited applicability	Utility maximization
Yao et al. [23]	ReAct	Pre-trained	No social grounding	Task success
Park et al. [1]	Generative	Fictional	High cost and scalability limits	User survey
Argyle et al. [5]	Silicon Sampling	ANES	Distribution mismatch	Correlation
This Work	Multi-Agent	Real Data	None	Quantitative/Distributional Metrics

Source: Own elaboration.

The comparative analysis in Table 1 highlights a consistent pattern across prior work. Symbolic architectures such as BDI [9] fail to scale beyond manually specified rules and therefore provide no mechanism for grounding agent behavior in empirical data. Reinforcement learning approaches [15] optimize reward functions, limiting their applicability to population-level simulation. ReAct-based agents [23] enable contextual reasoning and task interaction but still rely on pre-trained knowledge. Generative-agent systems [1] demonstrate emergent behavior but operate entirely within fictional settings, lacking any calibration to external data. Silicon Sampling [5] uses real datasets but models individuals as static samples, without multi-agent interaction, thereby preventing the emergence of collective dynamics.

This thesis addresses the limitations of all these approaches by combining real socioeconomic data, structured population synthesis [25], and a multi-agent architecture capable of controlled interaction. Instead of relying solely on pre-trained priors, agents are parameterized using empirical profiles that constrain their behavioral space. The resulting synthetic society is then evaluated using quantitative similarity metrics, including Wasserstein distance [35], to assess alignment between simulated and real-world distributions.



3

3

CONCLUSION

3.1 FINAL CONSIDERATIONS AND PLANNING

This monograph established the theoretical and conceptual foundations for the **Behavioral Grounding Framework (BGF)**, a methodological proposal designed to address the structural validity gap observed in contemporary synthetic societies powered by Large Language Models (LLMs). Throughout the literature review, we demonstrated that although LLM-based agents have substantially advanced beyond the rigidity of Symbolic architectures and the opacity of reinforcement learning systems, current approaches still lack mechanisms for empirical grounding, fidelity assessment, and behavioral consistency over time.

This study presents a conceptual framework for creating synthetic societies using socioeconomic records. By integrating population synthesis, generative reasoning, and interaction networks, the framework enables simulations that reflect demographic patterns and collective behaviors. Previous research shows that tools like the Wasserstein Distance and Gini Coefficient assess how closely simulated outcomes match real social data.

In this sense, the present work concludes the conceptual and theoretical stage of the research. The next phase will focus on the operationalization, experimentation, and empirical validation of the proposed framework, following the structured development plan presented below.

3.2 FUTURE WORK AND SCHEDULE

The subsequent stage of the project consists of transforming the conceptual foundations established into a functional simulation pipeline. To ensure methodological rigor and efficient progress, activities are structured to allow parallel execution where feasible. The planning encompasses theoretical refinement, system design, simulation execution, and the preparation of analytical components that will support the empirical validation of the Behavioral Grounding Framework.

The schedule below Table 2 details the roadmap for the first semester of 2026.

Table 2: Strategic Schedule for Research Development

Activity	Period
Phase 1: Foundations for Data Grounding Curation and preprocessing of socioeconomic microdata sources; Definition of stratification criteria and sampling logic for the generation of statistically representative agent personas.	Jan 2026
Phase 2: Cognitive Modeling and Interaction Design Modeling of the cognitive architecture (memory, reflection, and reasoning modules); Specification of interaction protocols and definition of the social network topology to govern agent exchanges.	Feb 2026
Phase 3: Simulation Execution and Calibration Configuration of the simulation environment and execution of pilot scenarios; Monitoring of emergent dynamics and calibration of behavioral parameters based on preliminary logs.	Mar 2026
Phase 4: Empirical Validation and Analytics Implementation of the statistical validation suite, including distributional distance metrics and inequality indicators; Comparative analysis between synthetic simulation outputs and empirical baselines.	Apr 2026
Phase 5: Synthesis and Documentation Consolidation of experimental results; Theoretical discussion of findings; Drafting of the final chapters covering architecture, implementation, and evaluation; Full revision of the manuscript.	May 2026
Phase 6: Final Submission and Defense Final formatting adjustments according to academic standards; Preparation of presentation materials; Formal submission and oral defense of the thesis.	Jun 2026

Source: Own elaboration.

This schedule ensures that theoretical consolidation, simulation design, and empirical evaluation progress cohesively while distributing computationally intensive and analytical tasks across the semester. The next phase is expected to solidify the Behavioral Grounding Framework as a foundational contribution to the study of synthetic societies and fidelity assessment in complex social systems.



REFERENCES

References

- [1] J. S. Park, J. O'Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein, "Generative agents: Interactive simulacra of human behavior," in *Proceedings of the 36th annual acm symposium on user interface software and technology*, 2023, pp. 1–22.
- [2] J. M. Epstein and R. Axtell, *Growing artificial societies: social science from the bottom up*. Brookings Institution Press, 1996.
- [3] N. Gilbert and K. Troitzsch, *Simulation for the social scientist*. McGraw-Hill Education (UK), 2005.
- [4] E. Bonabeau, "Agent-based modeling: Methods and techniques for simulating human systems," *Proceedings of the national academy of sciences*, vol. 99, no. suppl_3, pp. 7280–7287, 2002.
- [5] L. P. Argyle, E. C. Busby, N. Fulda, J. R. Gubler, C. Rytting, and D. Wingate, "Out of one, many: Using language models to simulate human samples," *Political Analysis*, vol. 31, no. 3, pp. 337–351, 2023.
- [6] J. J. Horton, "Large language models as simulated economic agents: What can we learn from homo silicus?" National Bureau of Economic Research, Tech. Rep., 2023.
- [7] P. Törnberg, "Best practices for text annotation with large language models," *arXiv preprint arXiv:2402.05129*, 2024.
- [8] M. Wooldridge, *An introduction to multiagent systems*. John wiley & sons, 2009.
- [9] A. S. Rao, M. P. Georgeff *et al.*, "Bdi agents: From theory to practice." in *Icmas*, vol. 95, 1995, pp. 312–319.
- [10] Wooldridge, *Reasoning about Rational Agents*. The MIT Press, Jul. 2000.
- [11] M. E. Bratman, D. J. Israel, and M. E. Pollack, "Plans and resource-bounded practical reasoning," *Computational intelligence*, vol. 4, no. 3, pp. 349–355, 1988.
- [12] M. P. Georgeff and A. L. Lansky, "Reactive reasoning and planning." in *AAAI*, vol. 87, 1987, pp. 677–682.
- [13] J. E. Laird, *The Soar cognitive architecture*. MIT press, 2019.
- [14] J. R. Anderson, *How can the human mind occur in the physical universe?* Oxford University Press, 2009.

- [15] R. S. Sutton, A. G. Barto *et al.*, *Reinforcement learning: An introduction*. MIT press Cambridge, 1998, vol. 1, no. 1.
- [16] P. Hernandez-Leal, B. Kartal, and M. E. Taylor, “A survey and critique of multi-agent deep reinforcement learning,” *Autonomous Agents and Multi-Agent Systems*, vol. 33, no. 6, pp. 750–797, 2019.
- [17] K. Zhang, Z. Yang, and T. Başar, “Multi-agent reinforcement learning: A selective overview of theories and algorithms,” *Handbook of reinforcement learning and control*, pp. 321–384, 2021.
- [18] S. Russell and P. Norvig, *Artificial intelligence: A modern approach, global edition*, 4th ed. Pearson Education, 2021.
- [19] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, “Attention is all you need,” *Advances in neural information processing systems*, vol. 30, 2017.
- [20] S. Bubeck, V. Chandrasekaran, R. Eldan, J. Gehrke, E. Horvitz, E. Kamar, P. Lee, Y. T. Lee, Y. Li, S. Lundberg *et al.*, “Sparks of artificial general intelligence: Early experiments with gpt-4,” *arXiv preprint arXiv:2303.12712*, 2023.
- [21] J. Wei, X. Wang, D. Schuurmans, M. Bosma, F. Xia, E. Chi, Q. V. Le, D. Zhou *et al.*, “Chain-of-thought prompting elicits reasoning in large language models,” *Advances in neural information processing systems*, vol. 35, pp. 24 824–24 837, 2022.
- [22] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel *et al.*, “Retrieval-augmented generation for knowledge-intensive nlp tasks,” *Advances in neural information processing systems*, vol. 33, pp. 9459–9474, 2020.
- [23] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. R. Narasimhan, and Y. Cao, “React: Synergizing reasoning and acting in language models,” in *The eleventh international conference on learning representations*, 2022.
- [24] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao, “React: Synergizing reasoning and acting in language models,” 2022, project Website. [Online]. Available: <https://react-lm.github.io/>
- [25] R. J. Beckman, K. A. Baggerly, and M. D. McKay, “Creating synthetic baseline populations,” *Transportation Research Part A: Policy and Practice*, vol. 30, no. 6, pp. 415–429, 1996.

- [26] D. J. Watts and S. H. Strogatz, “Collective dynamics of ‘small-world’ networks,” *nature*, vol. 393, no. 6684, pp. 440–442, 1998.
- [27] J. S. Park, C. Q. Zou, A. Shaw, B. M. Hill, C. Cai, M. R. Morris, R. Willer, P. Liang, and M. S. Bernstein, “Generative agent simulations of 1,000 people,” *arXiv preprint arXiv:2411.10109*, 2024.
- [28] G. V. Aher, R. I. Arriaga, and A. T. Kalai, “Using large language models to simulate multiple humans and replicate human subject studies,” in *International conference on machine learning*. PMLR, 2023, pp. 337–371.
- [29] A. Li, H. Chen, H. Namkoong, and T. Peng, “Llm generated persona is a promise with a catch,” *arXiv preprint arXiv:2503.16527*, 2025.
- [30] J. Bisbee, J. D. Clinton, C. Dorff, B. Kenkel, and J. M. Larson, “Synthetic replacements for human survey data? the perils of large language models,” *Political Analysis*, vol. 32, no. 4, pp. 401–416, 2024.
- [31] Y. Gao, D. Lee, G. Burtch, and S. Fazelpour, “Take caution in using llms as human surrogates: Scylla ex machina,” *arXiv preprint arXiv:2410.19599*, 2024.
- [32] A. Collins, M. Koehler, and C. Lynch, “Methods that support the validation of agent-based models: An overview and discussion,” *Journal of Artificial Societies and Social Simulation*, vol. 27, no. 1, 2024.
- [33] D. J. MacKay, *Information theory, inference and learning algorithms*. Cambridge university press, 2003.
- [34] M. Arjovsky, S. Chintala, and L. Bottou, “Wasserstein generative adversarial networks,” in *International conference on machine learning*. PMLR, 2017, pp. 214–223.
- [35] S. Manzan and L. Wang, “Estimation of agent-based models using the wasserstein distance,” *Available at SSRN 5284325*, 2024.
- [36] F. Cowell, *Measuring inequality*. Oxford University Press, 2011.



idp Ensino que
te conecta